

Macroeconomics

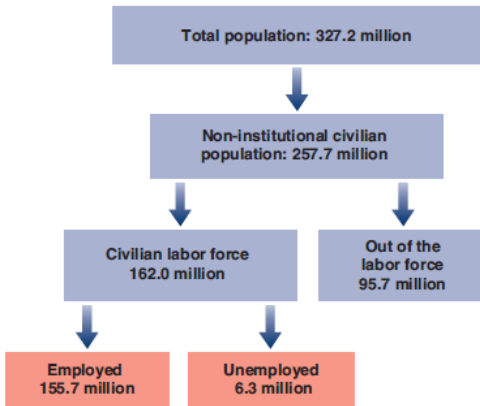
Labor Market

(Blanchard, Macroeconomics 8th edition Ch. 7-8)

Labor Market

- ▶ We have focused on the short run by assuming a constant price level in the IS-LM model.
- ▶ We now turn to the medium run and explore how prices and wages adjust over time, and how this in turn affects output.
- ▶ The labor market is the center of that sequence of events.

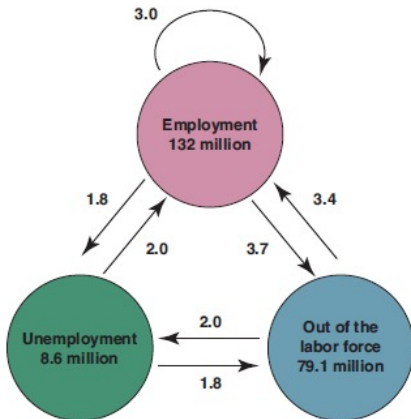
US Labor Market, 2018



Labor Market

- ▶ A given unemployment rate may reflect either:
 - ▶ An active labor market: Many separations and hires, i.e, many workers entering and exiting unemployment
 - ▶ A sclerotic labor market: Few separations and hires, and a stagnant unemployment pool
- ▶ The US Current Population Survey (CPS) shows the average monthly flows.
- ▶ Separations include quits and layoffs.

US Average Monthly Flows, 1994 to 2018 (millions)



US Average Monthly Flows, 1994 to 2014

1. The flows of workers in and out of employment are large.
 - ▶ On average, there are 8.5 million separations (quits and layoffs) each month in US.
 - ▶ Continual job destruction and job creation across firms.
2. The flows in and out of unemployment are large relative to the number of unemployed.
 - ▶ The average duration of unemployment (the length of time people spend unemployed) is 2 to 3 months.
 - ▶ In many European countries, the average duration is much longer than in US.
 - ▶ When unemployment is high (e.g., crisis), the average duration of unemployment becomes much longer.
3. There are also large flows in and out of the labor force, much of it directly to and from employment.

Discouraged Workers

- ▶ Each month only about 450,000 new people (e.g., students) enter the labor force, and about 350,000 retire.
- ▶ Many who are classified as “out of the labor force” are in fact **discouraged workers**: not actively looking for a job but will take it if they find one.
- ▶ So, rather than the unemployment rate, economists sometimes focus on the employment rate: the ratio of employment to the population.

The Australian Monthly Population Survey (MPS)

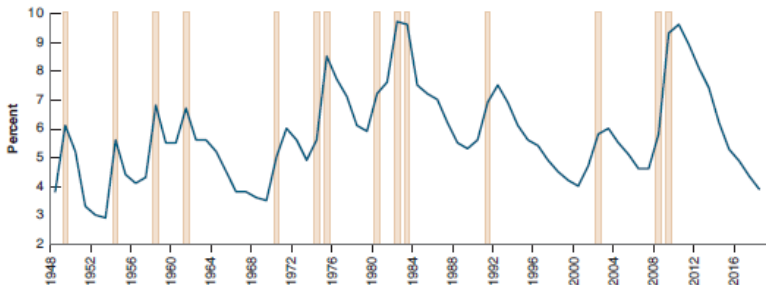
- ▶ Gathers varied information about the country's labor force. One of the main objectives of the survey is to measure changes in the households over time.
- ▶ The sample size used has increased from 30,000 to 300,000 households, and replaced every eight months.
- ▶ Around 43,000 permanent and temporary census staff members are employed to collect data, conducting all interviews either online or face-to-face.
- ▶ Census data are used in compiling the monthly national statistics on employment, unemployment, gender, size, and earnings of households.
- ▶ It also counts individuals in each geographic area to allocate Commonwealth funds to state and local governments.

The Australian Monthly Population Survey (con.)

- ▶ Economic analysts and policy makers use the data both in its static and continuous forms.
- ▶ Static: a snapshot of certain aspects at one point in time is collected and compared with other snapshots collected over previous years
 - ▶ e.g., change in the number of hours worked in each industry over the past 10 years.
- ▶ Dynamic: census workers compare the development of the situation of the same household over time
 - ▶ e.g., changes in conditions of employment or the patterns of household spending on a wide range of goods and services.
- ▶ Denmark: <https://www.dst.dk/en/Statistik/emner/arbejde-og-indkomst/befolkningens-arbejdsmarkedssstatus>

Movements in US Unemployment, 1948-2018

- ▶ Since 1948, the average yearly U.S. unemployment rate has fluctuated between 3 and 10%.
- ▶ Shaded area = recession



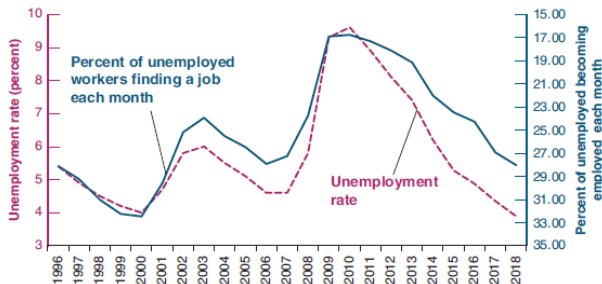
Source: Series UNRATE: Federal Reserve Economic Data (FRED)

<http://research.stlouisfed.org/fred2/> .

Movements in Unemployment

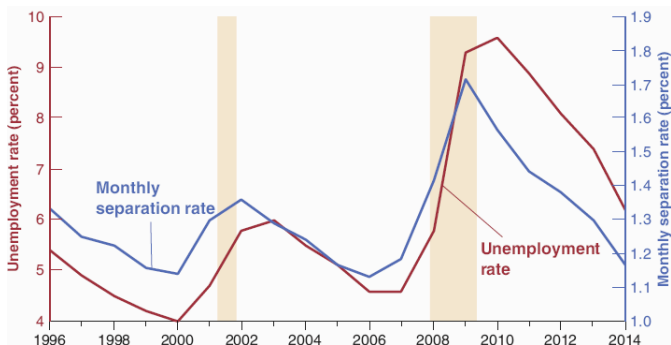
- ▶ How can firms decrease their employment in response to a decrease in demand?
- ▶ They can hire fewer new workers, or they can lay off the workers they currently employ.
- ▶ Typically, firms prefer to slow or stop the hiring of new workers first, relying on quits and retirements to achieve a decrease in employment, then have to lay off workers.
- ▶ What are the implications for employed and unemployed workers?
 - ▶ Fewer hires means that the chance that an unemployed worker will find a job diminishes.
 - ▶ Higher layoffs means that employed workers are at a higher risk of losing their job.

Proportion of Unemployed Finding Jobs



- ▶ When unemployment is higher, the proportion of unemployed becoming employed within one month is lower.
- ▶ Note that the scale on the right is an inverted scale.

Monthly Separation Rate from Employment



- When unemployment is higher, a higher proportion of workers become unemployed.

Summary

- ▶ When unemployment is high, workers are worse off in two ways:
 1. Employed workers face a higher probability of losing their job.
 2. Unemployed workers face a lower probability of finding a job; or they can expect to remain unemployed for a longer time.

Wage Determination

- ▶ Sometimes wages are set by collective bargaining: a bargaining between unions and firms.
- ▶ Slightly more than 10% of U.S. workers' wages are set by collective bargaining.
- ▶ For the rest, wages are either set by employers or by bargaining between the employer and individual employees.
- ▶ The higher the skills needed to do the job, the more likely there is to be bargaining between employers and individual employees, e.g., LeBron James
- ▶ Collective bargaining plays an important role in Japan and most European countries:
<https://www.youtube.com/watch?v=ygbXzNWBOSs>

Collective Bargaining

Country	% Covered by collective bargaining
France	98%
Finland	91%
Sweden	89%
Denmark	80%
Spain	69%
Germany	59%
Ireland	44%
Latvia	34%
UK	29%
Greece	10%
EU average	60%

Collective Bargaining (con.)

- ▶ High collective bargaining coverage may go with high union density (e.g., Sweden, Denmark), or could reflect legal framework (e.g., Austria, France)
- ▶ In France, almost all employees are covered by industry-level agreements, but these often provide only minimum terms, with improvements negotiated at company level or individually with employees.
- ▶ In Sweden and Denmark, although most employees are covered by industry deals, lower-level negotiators have substantial room for manoeuvre.
- ▶ In Sweden, only 10% of all employees had their pay entirely set by industry-level collective agreements in the 2013 bargaining round.
- ▶ In Denmark, in the biggest group of agreements, covering 85% of employees in labor unions in the private sector, pay is set by local deals at company level, i.e., important parts of the pay package are left to negotiations at the local level.

Bargaining Power

- ▶ Workers are typically paid a wage exceeding their **reservation wage**: the wage that would make them indifferent between working or being unemployed.
- ▶ Wages typically depends on labor-market conditions: The lower the unemployment rate, the higher the wages.
- ▶ Workers' bargaining power depends on:
 - ▶ How costly for the firm to find other workers, e.g., McDonalds vs. surgeon
 - ▶ How hard for workers to find another job if they were to leave the firm, e.g., depends on unemployment rate, labor-market conditions

Efficiency Wages

- ▶ Efficiency wage theories link the productivity or the efficiency of workers to the wage they are paid.
- ▶ Firms may want to pay a wage above the reservation wage in order to decrease workers' turnover and increase productivity.
- ▶ Firms that see employee morale and commitment as essential to the quality of workers' work will pay more than those whose activities are routine, e.g., Google
- ▶ When unemployment is low, firms that want to avoid an increase in quits will increase wages to induce workers to stay with the firms.

Henry Ford's Model T



Henry Ford and Efficiency Wages

- ▶ In 1914, Ford announced to pay all qualified employees min. \$5 a day for an 8 hr day, compared to previous ave. \$2.30 for a 9 hr day.
- ▶ What Ford's motivations were is not entirely clear. The reason was not that the company had a hard time finding workers at the previous wage. But the company clearly had a hard time retaining workers (high turnover rate, dissatisfaction)

	1913	1914	1915
Turnover rate (%)	370	54	16
Layoff rate (%)	62	7	0.1

Source: Dan Raff and Lawrence Summers, "Did Henry Ford Pay Efficiency Wages?" *Journal of Labor Economics* 1987 5 (No. 4 Part 2): pp. S57–S87.

- ▶ Productivity and profits increased; because of Model-T's success?
- ▶ Other objectives?: keeping unions out, generate publicity (e.g., Amazon and minimum wage \$15)

Wage Determination

- ▶ Aggregate nominal wage W :

$$\ominus \oplus$$
$$W = P^e F(u, z)$$

- ▶ Workers and firms care about **real wages** (W/P), not nominal.
- ▶ The **nominal wage** depends on the expected price level (rather than the actual price level) because when nominal wages are set, the relevant price levels are not yet known.
- ▶ e.g., in some union contracts, nominal wages are set in advance for three years. If the price level goes up unexpectedly during the year, nominal wages are typically not readjusted.

Wage Determination (con.)

- ▶ An increase in the unemployment rate u decreases wages.
- ▶ Higher unemployment either weakens worker' bargaining power, or allows firms to pay lower wages and still keep workers willing to work.
- ▶ z stands for all the factors that affect wages given the expected price level and the unemployment rate, for example:
 - ▶ Unemployment insurance as the payment of unemployment benefits to workers who lose their jobs.
 - ▶ Employment protection makes it more expensive to lay off workers.

Price Determination

- ▶ The prices set by firms depends on their costs, which in turn depends on the nature of the **production function**:

$$Y = AN$$

- ▶ where Y is output, N is employment and A is labor productivity (output per worker).
- ▶ The production function is the relation between the inputs used in production and the quantity of output produced, and on the prices of these inputs.

Price Determination (con.)

- ▶ Assume that A is constant and $A = 1$, then:

$$Y = N$$

- ▶ which implies that the cost of producing one more unit of output is the cost of employing one more worker at W .
- ▶ i.e., the marginal cost of production is equal to W .
- ▶ Now assume firms set their price according to a markup m over the cost so that:

$$P = (1 + m)W$$

The Natural Rate of Unemployment

- ▶ Assume that W depends on the actual price level (P) rather than the expected price level (P^e) (i.e., assume $P^e = P$)

$$\ominus \oplus$$
$$W/P = F(u, z)$$

- ▶ The higher the unemployment rate, the lower the real wage chosen by wage setters (weaker bargaining position).
- ▶ The **wage-setting relation** is the relation between the real wage and the rate of unemployment.

Price-Setting Relation

- ▶ Now divide both sides of the price-determination equation by the nominal wage:

$$\frac{P}{W} = 1 + m$$

- ▶ Inverting both sides gives the implied real wage, or the **price-setting relation**:

$$\frac{W}{P} = \frac{1}{1 + m}$$

- ▶ Price-setting decisions determine the real wage paid by firms.

Equilibrium Real Wages and Unemployment

- ▶ The equilibrium unemployment rate is determined by

$$F(u_n, z) = \frac{1}{1 + m}$$

- ▶ u_n depends on z and m .
- ▶ u_n is also called the **natural rate of unemployment**,
- ▶ or the **structural rate of unemployment**: factors like the generosity of unemployment benefits or antitrust legislation can hardly be thought of as the result of nature.
- ▶ Rather, they reflect various characteristics of the structure of the economy.

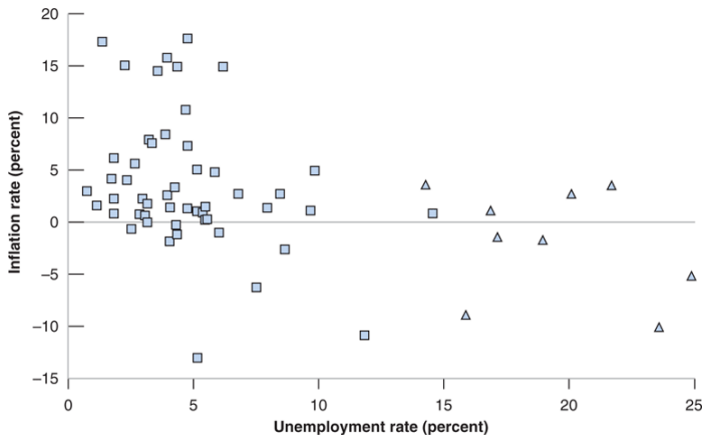
Equilibrium Real Wages and Unemployment

Unemployment Benefits

The Phillips Curve

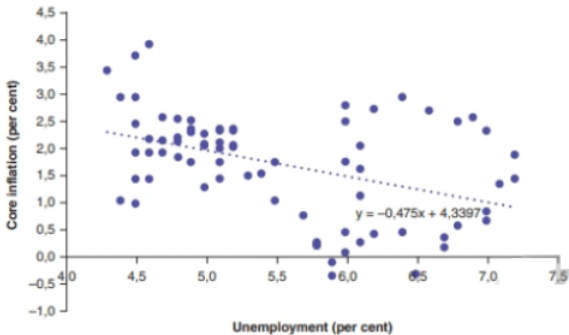
- ▶ In 1958, A.W. Phillips found a negative relation between inflation and unemployment.
 - ▶ When unemployment was low, inflation was high.
 - ▶ When unemployment was high, inflation was low, often even negative.
- ▶ Two years later, Paul Samuelson and Robert Solow labeled this relation the *Phillips curve*, which became central to macroeconomic thinking and policy.
- ▶ It appeared that a country could achieve low unemployment if it were willing to tolerate higher inflation.
- ▶ Or it could achieve price level stability (zero inflation) if it were willing to tolerate higher unemployment.

Inflation vs Unemployment (US 1900-1960)



Source: Historical Statistics of
the United States. [http://hsus.
cambridge.org/HSUSWeb/index.do](http://hsus.cambridge.org/HSUSWeb/index.do)

Phillips Curve in Europe



Inflation, Expected Inflation, and Unemployment

- ▶ Recall the wage determination equation:

$$W = P^e F(u, z)$$

- ▶ And the price determination equation:

$$P = (1 + m)W$$

- ▶ Substitute for W :

$$P = P^e(1 + m)F(u, z)$$

Inflation, Expected Inflation, and Unemployment

Interpretation

- ▶ An increase in $\pi^e \Rightarrow$ increase in π .
- ▶ Given π^e , an increase in m , or an increase in $z \Rightarrow$ increase in π .
- ▶ Given π^e , a decrease in $u \Rightarrow$ increase in π .
- ▶ Adding time indices:

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

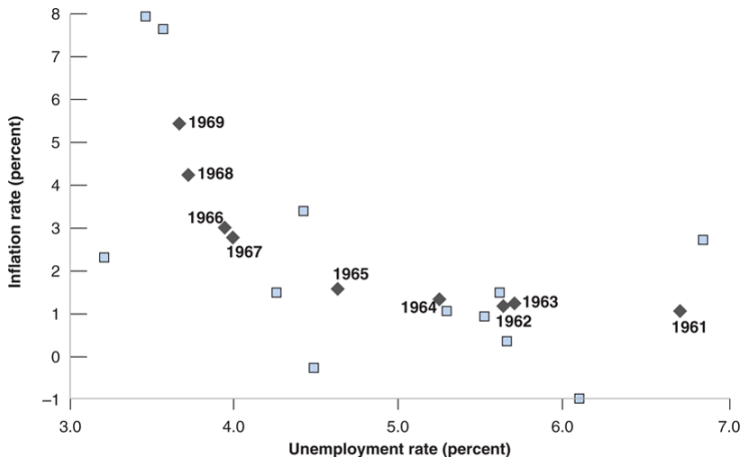
The Phillips Curve and Its Mutations

- ▶ Assume $\pi_t^e = \bar{\pi}$, so

$$\pi_t = \bar{\pi} + (m + z) - \alpha u_t$$

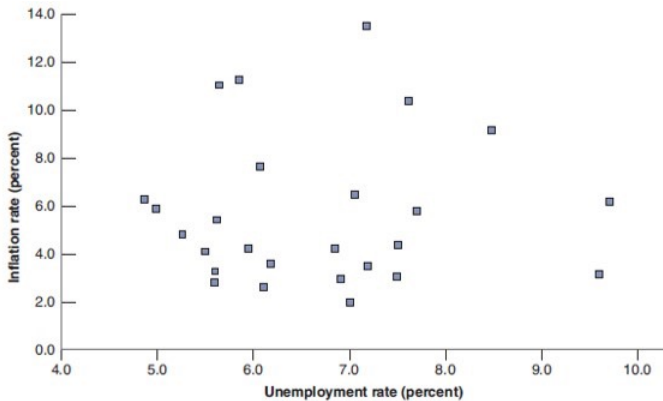
- ▶ which is a negative relation between unemployment and inflation.
- ▶ This Phillips curve relation was observed by Phillips for the UK, and Solow and Samuelson for the US in the 1960s.
- ▶ The steady decline in the US unemployment rate throughout the 1960s was associated with a steady increase in the inflation rate.
- ▶ The relation vanished in the 1970s because wage setters changed the way they formed inflation expectations.

Inflation versus Unemployment (US 1948-1969)



Source: Series UNRATE,
CPIAUSCL Federal Reserve Eco-
nomic Data (FRED) <http://research.stlouisfed.org/fred2/>

Inflation versus Unemployment (US 1970-1995)



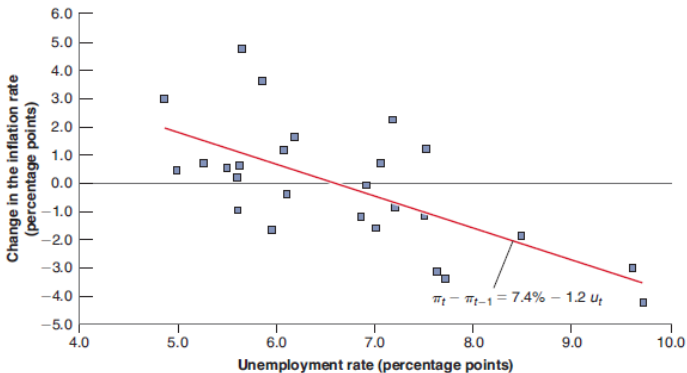
Change in Expectations

- ▶ The rate of inflation became more persistent, and people, when forming expectations, started to take into account this persistence.
- ▶ Suppose expected inflation this year depends on a constant value $\bar{\pi}$ with weight $1 - \theta$, and partly on inflation last year with weight θ :

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

- ▶ When $\theta = 0$: $\pi_t = \bar{\pi} + (m + z) - \alpha u_t$
- ▶ When $\theta > 0$: $\pi_t = [(1 - \theta)\bar{\pi} + (m + z)] + \theta\pi_{t-1} - \alpha u_t$
- ▶ When $\theta = 1$: $\pi_t - \pi_{t-1} = (m + z) - \alpha u_t$

Change in Inflation vs Unemployment (US 1970-1995)



Phillips Curve

- ▶ The line that best fits the scatter of points is:

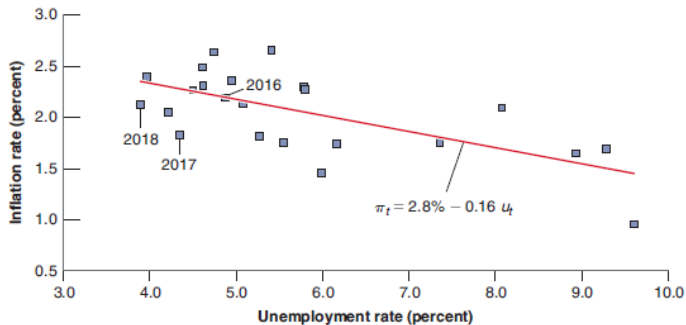
$$\pi_t - \pi_{t-1} = 7.4\% - 1.2u_t$$

- ▶ “Modified Phillips curve”, aka expectations-augmented Phillips curve, aka accelerationist Phillips curve.
- ▶ We will call this the “Phillips curve”, as opposed to the “original Phillips curve”.

The Re-anchoring of Expectations

- ▶ By the mid-1990s the Fed had largely achieved its goal of keeping inflation around 2%.
- ▶ Even if inflation was higher than the target in a given year, people assumed that the central bank would take measures to return inflation to its target value in the future.
- ▶ Expected inflation became roughly constant, equal to the target inflation set by the central bank.
- ▶ Stable inflation changed the way people formed expectations yet again.
- ▶ Expectations of inflation that became de-anchored during the 1970s and 1980s became re-anchored in the mid-1990s.

Inflation vs Unemployment (US 1996-2018)



Phillips Curve and the Natural Rate of Unemployment

- ▶ Natural rate of unemployment: unemployment rate at which the actual price level is equal to the expected price level.
- ▶ Which implies that the actual inflation rate is equal to the expected inflation rate.
- ▶ Substitute $\pi = \pi^e$ and solve for u_n :

$$u_n = \frac{m + z}{\alpha}$$

- ▶ Rewrite the difference between actual inflation and expected inflation as:

$$\begin{aligned}\pi_t - \pi_t^e &= -\alpha \left(u_t - \frac{m + z}{\alpha} \right) \\ \Rightarrow \pi_t - \pi_t^e &= -\alpha (u_t - u_n)\end{aligned}$$

Natural Rate of Unemployment

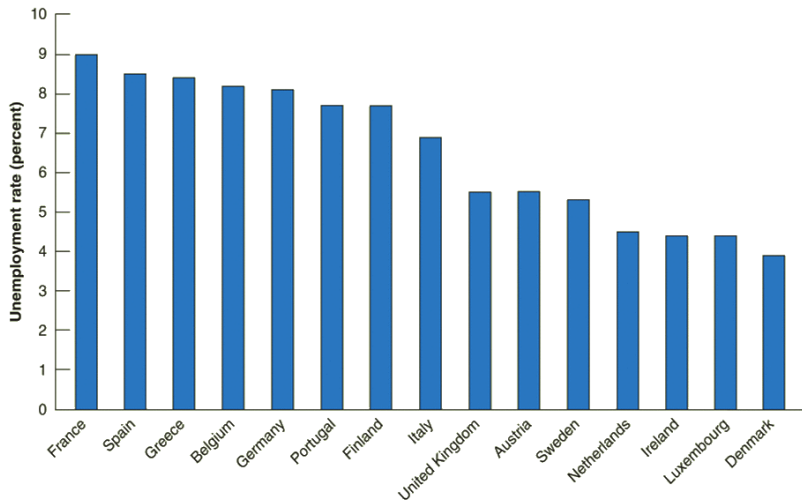
- ▶ Is there a “natural rate” of unemployment? Or can any level be achieved as long as policymakers tolerate, e.g., higher inflation?
- ▶ In equilibrium, P need not be equal to P^e , and π need not be equal to π^e .
- ▶ However, an unemployment rate consistently lower than u_n requires wage setters to systematically underpredict inflation.
- ▶ Milton Friedman and Edmund Phelps also argued that it is not possible to sustain lower unemployment with higher inflation, i.e., temporary tradeoff.
- ▶ u that is different from u_n may lead to accelerating inflation or deflation (as opposed to steady/stable inflation).

Examples

Variations Across Countries

- ▶ The natural rate of unemployment depends on all the factors that affect wage setting (z, m, α).
- ▶ If these factors differ across countries, there is no reason to expect all countries to have the same natural rate of unemployment.
- ▶ e.g., unemployment rate in the Euro area has averaged close to 9% since 1990.
- ▶ A high unemployment rate for a few years may well reflect a deviation of the unemployment rate from the natural rate.
- ▶ A high average unemployment rate for 25 years, together with no sustained decrease in inflation, surely reflects a high natural rate.

Unemployment Rates in 15 European Countries, 2006



What Explains European Unemployment?

- ▶ Factors for labor-market rigidities:
 - ▶ A generous system of unemployment insurance
 - ▶ A high degree of employment protection (severance payment)
 - ▶ Minimum wages
 - ▶ Bargaining rules, such as extension agreements
 - ▶ <https://www.youtube.com/watch?v=4jrtWjlpXr4>,
<https://www.youtube.com/watch?v=fCiXxbEQ81g>
- ▶ Unemployment was not always high: 2-3% in 1960s
- ▶ Social protection had been high before, increased in 1970s after shocks to price of oil

What Explains European Unemployment?

- ▶ Countries such as Ireland or the UK indeed have labor-market institutions that resemble those of the US:
 - ▶ Limited benefits, low employment protection, and weak unions.
- ▶ But countries such as Denmark or the Netherlands have a high degree of social protection (in particular high unemployment benefits) and strong unions.
- ▶ Employment protection (e.g., generous severance payments) may be consistent with low unemployment, so long as firms do not face the prospect of long administrative or judicial uncertainty when they lay off workers.

Low Unemployment Rate in Japan

- ▶ Official unemployment rate in 2016: only 3.1%
- ▶ Natural rate of unemployment in 2015: 2.25%, far below observed figures in other industrial nations (e.g., 5.5% in US).
- ▶ Yet, the low unemployment rate is not a sign of economic recovery from the long-lived Japanese economic contraction, but is reason for concern as it is indicative of a tight labor market.
- ▶ Labor shortages have prompted many Japanese housewives and retirees to enter the labor market
 - ▶ Raises questions about the capability of these re-entrants to keep up with technological advancements and managerial innovations.

Possible Explanations

- ▶ Declining bargaining power of trade unions:
 - ▶ Shunto: springtime annual Japanese ritual of wage bargaining between business groups and labor unions.
 - ▶ Trade union membership has dropped substantially: rise in number of temporary employees (ineligible to participate in labor unions).
- ▶ Japanese seniority-based wage system: nenko joretsu
 - ▶ While the system underpays young workers, it rewards them well in later years awarding an increase in pay for each year of service.
 - ▶ Strong incentive to remain with their first employer, hence providing de facto lifetime employment.
 - ▶ Also an early retirement age that forces Japanese employees to retire at the ages of 55 to 62.
 - ▶ Younger generations have the chance to acquire jobs, reducing unemployment levels.

Possible Explanations (con.)

- ▶ Shrinking population and high longevity:
 - ▶ Childbearing level in Japan is among the lowest in the world, resulting in a decline of almost one million people every year.
 - ▶ In addition, second highest longevity in the world.
 - ▶ Combined effect: for every retiree there are three working people.
- ▶ Flexible remuneration system:
 - ▶ Low basic wages and high overtime pay and higher bonuses.
 - ▶ Allows businesses to retain workers as they further cut bonuses and overtime pay when the economy slumps.
 - ▶ Double-edged sword: As workers take home a lower pay, it becomes more difficult for the Japanese economy to recover from its persistent deflationary slump
- ▶ Sectoral factors:
 - ▶ Employment in agricultural sector: 3.7% of total employment.
 - ▶ Much higher than the UK, Belgium, and Germany (1%).

Wage Indexation

- ▶ When the inflation rate becomes high, the terms of wage agreements tend to change with the level of inflation.
- ▶ Wage indexation is a provision that automatically increases wages in line with inflation.
- ▶ Suppose proportion λ of labor contracts is indexed, so nominal wages move one-for-one with changes in the actual price level, i.e.,

$$W = [\lambda P + (1 - \lambda)P^e]F(u, z)$$

- ▶ Further assume for simplicity that $\pi_t^e = \pi_{t-1}$, then:

$$\pi_t = [\lambda \pi_t + (1 - \lambda)\pi_{t-1}] - \alpha(u_t - u_n)$$

Wage Indexation (con.)

- ▶ When $\lambda = 0$: $\pi_t - \pi_{t-1} = -\alpha(u_t - u_n)$
- ▶ When $\lambda > 0$:

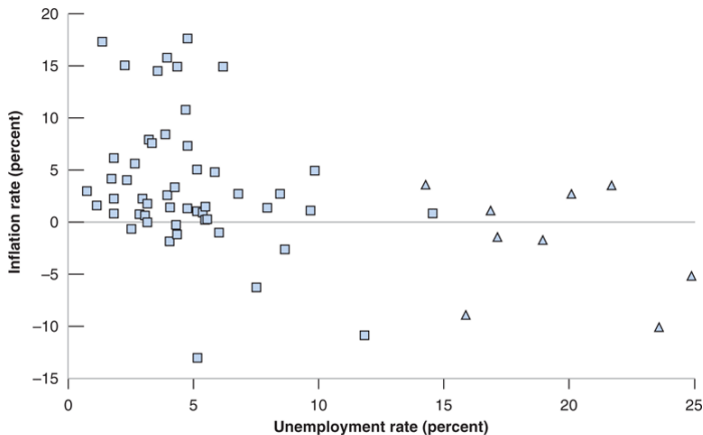
$$\pi_t - \pi_{t-1} = -\frac{\alpha}{(1 - \lambda)}(u_t - u_n)$$

- ▶ Without wage indexation, lower unemployment increases wages, which in turn increases prices.
- ▶ But because wages do not respond to prices right away, there is no further increase in prices within the year.
- ▶ Wage indexation increases the effect of unemployment on inflation.
- ▶ https://www.youtube.com/watch?v=V_u3bzET1LQ

Wage Indexation (con.)

- ▶ With wage indexation, an increase in prices leads to a further increase in wages within the year.
- ▶ This leads to a further increase in prices, and so on, so that the effect of unemployment on inflation within the year is higher.
- ▶ If, and when, λ gets close to 1 (i.e., most labor contracts allow for wage indexation) small changes in unemployment can lead to large changes in inflation.
- ▶ That is, there can be large changes in inflation with nearly no change in unemployment.
- ▶ This is what happens in countries where inflation is high.
- ▶ The relation between inflation and unemployment becomes more and more tenuous and eventually disappears altogether.

Inflation vs Unemployment (US 1900-1960)



Source: Historical Statistics of
the United States. [http://hsus.
cambridge.org/HSUSWeb/index.do](http://hsus.cambridge.org/HSUSWeb/index.do)

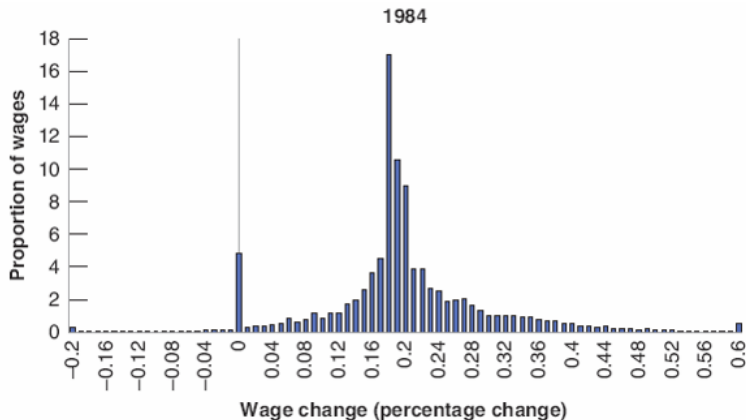
Deflation

- ▶ Points in 1930s (triangles) lie to the right
- ▶ Unemployment is high (Great Depression), *and* inflation rate is surprisingly high
- ▶ Given the high unemployment rate, we would have expected not merely deflation, but a large rate of deflation
- ▶ How do we interpret this fact?

Deflation (con.)

1. Increase not only in the actual unemployment rate but also in the natural unemployment rate:
 - ▶ But, most economic historians see the Great Depression primarily as the result of a large adverse shift in aggregate demand.
 - ▶ This led to an increase in the actual unemployment rate *over* the natural rate of unemployment.
2. When the economy starts experiencing deflation, the Phillips curve relation breaks down.
 - ▶ One possible reason: reluctance of workers to accept decreases in their nominal wages.
 - ▶ e.g., Portugal, in 1984, the inflation rate was 27%, and the distribution of wage wages was roughly symmetric.
 - ▶ In 2012, the inflation rate was just 2.1%, and the distribution of wages was bunched at zero with nearly no negative wage changes.

Distribution of wage changes in Portugal, 1984



Distribution of wage changes in Portugal, 2012

